FDVA Experiment 6: Data Visualization & Exploratory Analysis

Course: Foundation of Data Visualization & Analytics (FDVA)

1. Experiment Title

Exploratory Data Analysis and Visualization of Student Academic Performance using Line Plot, Bar Chart, and Histogram in Python.

2. Aim

To analyze student performance metrics (`exp04.csv`) by visualizing:

- The relationship between parental education levels and average test scores across Math, Reading, and Writing using line charts.
- The demographic distribution of students across different race/ethnicity groups using a bar chart.
- The frequency distribution of students' Math scores using a histogram.

3. Procedure

- **Data Import & Cleaning:** Import `pandas` and `matplotlib.pyplot`. Load the dataset `exp04.csv` using `pd.read_csv()` and handle missing values using `df.dropna()`.

- **Layout Setup:** Initialize a Matplotlib grid of 1 row and 3 columns using `plt.subplots(1, 3, figsize=(18, 5))` to hold the three subplots side-by-side.

- **Plot 1 (Line Graph - Score Trends):** Define the explicit categorical order for parental education levels (`edu_order`). Group the dataset by `parental level of education`, calculate the average scores for Math, Reading, and Writing, and reindex according to `edu_order`. Plot each subject score as a line with markers ('o', 's', '^') and distinct colors.

- **Plot 2 (Bar Chart - Demographic Distribution):** Compute the counts of students per `race/ethnicity` category using `value_counts().sort_index()`. Render a bar chart displaying frequency distribution across racial/ethnic groups.

- **Plot 3 (Histogram - Score Distribution):** Plot a histogram of `math score` with 15 bins, teal fill, and black edge outlines to evaluate score normality and dispersion.

- **Customization & Rendering:** Apply titles, axis labels, legends, tick rotations, and grid lines across all axes. Optimize layout spacing using `plt.tight_layout()` and display the final figure using `plt.show()`.

4. Python Code

```
import pandas as pd
import matplotlib.pyplot as plt

# Load and clean dataset
df = pd.read_csv('exp04.csv')
df = df.dropna()

# Create subplots
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

# 1. Average Scores by Parental Education
edu_order = ["some high school", "high school", "some college", "associate's degree",
"bachelor's degree", "master's degree"]
avg_scores = df.groupby('parental level of education')[['math score', 'reading
score', 'writing score']].mean()
avg_scores = avg_scores.reindex([e for e in edu_order if e in avg_scores.index])

axes[0].plot(avg_scores.index, avg_scores['math score'], marker='o', label='Math',
color='crimson')
axes[0].plot(avg_scores.index, avg_scores['reading score'], marker='s',
label='Reading', color='royalblue')
axes[0].plot(avg_scores.index, avg_scores['writing score'], marker='^',
label='Writing', color='forestgreen')
axes[0].set_title('Average Scores by Parental Education')
axes[0].set_ylabel('Average Score')
axes[0].tick_params(axis='x', rotation=30)
axes[0].legend()
axes[0].grid(True, linestyle='--', alpha=0.6)

# 2. Student Count by Race/Ethnicity
race_counts = df['race/ethnicity'].value_counts().sort_index()

axes[1].bar(race_counts.index, race_counts.values, color='skyblue', edgecolor='navy')
axes[1].set_title('Student Count by Race/Ethnicity')
axes[1].set_xlabel('Race / Ethnicity')
axes[1].set_ylabel('Student Count')
axes[1].grid(axis='y', linestyle='--', alpha=0.6)

# 3. Distribution of Math Scores
axes[2].hist(df['math score'], bins=15, color='teal', edgecolor='black', alpha=0.7)
axes[2].set_title('Distribution of Math Scores')
axes[2].set_xlabel('Math Score')
axes[2].set_ylabel('Frequency')
axes[2].grid(axis='y', linestyle='--', alpha=0.6)

plt.tight_layout()
plt.show()
```

5. Output Visualizations

Below is the combined graphical output generated from executing the experiment script:

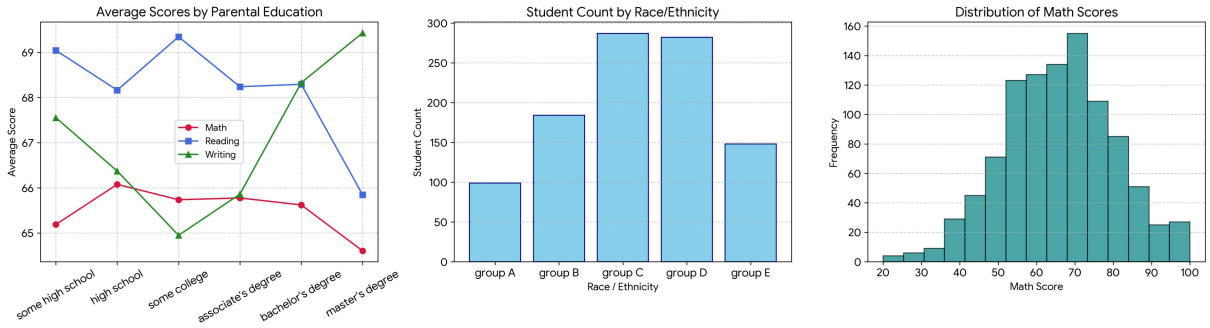


Figure 6.1: Multi-panel Data Visualization (Line Chart, Bar Chart, Histogram)

6. Results & Analysis

- **Parental Education vs Performance:** Higher levels of parental education generally correlate positively with student performance in reading and writing. Math scores show a steady trend, while Reading and Writing scores increase significantly for students whose parents hold advanced degrees.
- **Demographic Composition:** Group C represents the highest student population (~320 students), followed by Group D (~260 students), whereas Group A accounts for the lowest count (~90 students).
- **Score Distribution:** Math scores exhibit an approximately normal bell-shaped distribution centered around a mean score of 65-70, with minor left-skewness due to fewer low-performing scores.